Role of Umbilical Cord Blood Transplantation

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64.1 Introduction

- Umbilical cord blood transplant (UCBT) from unrelated donors (UD) was developed to overcome the limitations of HLA-matched bone marrow (BM)/peripheral blood (PB) UDs and improve access to allogeneic transplant (allo-HCT) both by increasing the donor pool and fasten time to HCT.
- UCBT activity has sharply decreased in favor of partially HLA-matched related donor transplants (haplo-HCT) using posttransplant cyclophosphamide (Passweg et al. 2021). This procedure shares most of the advantages of UCBT. It has expanded its use worldwide due to its effectiveness, lower financial impact, and relative simplicity. Two randomized trials have shown superiority of haplo-HCT over UCBT (Fuchs et al. 2020; Sanz et al. 2020). However, it is still a clinical option for some patients lacking other alternatives, due to either the absence of relative donors at due time or the presence of donor-specific antibodies. Whether or not it could be prioritized over haplo donors with negative characteristics like older age or gender mismatch should be the focus of future research.
- Recent advances in conditioning, graft manipulation, and supportive care have improved UCBT outcomes. These advances reflect in very encouraging outcomes after UCBT for adults and children with high-risk leukemias and some genetic diseases.

64.2 Potential Advantages and Disadvantages of UCBT

UCBT versus BMT/PBHCT	
Advantages	Disadvantages
• Expanded access to transplant* – Higher availability of donor* – Faster search and shorter time to transplant* – Greater HLA disparity allowed with low incidence of GVHD* • Lower risk of transmission of viral infections • More versatile transplant planning* • No risk of donor refusal • No risk to the donor	• Slower engraftment • Higher risk of non-immunological rejection (graft failure) • Remote possibility of transmission of a genetic disease* • Greater delay in immune reconstitution • No possibility of donor lymphocyte infusion*
*Advantages shared with partially matched related (haploidentical) HCT	*Disadvantages not shared with haploidentical HCT

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Similar to UCBT, haploidentical HCT can also be used on an urgent basis and extends donor availability to the vast majority of patients. In addition, haploidentical HCT allows a DLI if necessary.

64.3 Indications

Except for some patients with severe bone marrow failure, such as aplastic anemia and paroxysmal nocturnal hemoglobinuria, UCBT in adults is performed almost exclusively in patients with malignant hematological diseases. However, UCBT in children has been used for many other nonmalignant diseases, including primary immunodeficiency diseases and inherited metabolic disorders.

The American Society for Blood and Marrow Transplantation (ASBMT) (Kanate et al. 2020) and the European Society for Blood and Marrow Transplantation (EBMT) (Snowden et al. 2022) have recently published their respective guidelines that include recommendations for transplant indications in children and adults. It should be noted that the ASBMT did not differentiate recommendations for transplant indications based on donor or graft source and EBMT grouped the same indications for all HLA-mismatched alternative donors (UCB, haplo, and mismatched unrelated donors).

Allo-HCT indications should not defer according to donor or graft source but rather the expertise of the transplant center along with patient- and/or disease-related factors, and availability should determine the donor/graft source for the individual patient. More recent data shows that UCBT is probably a preferred donor choice for patients with very high-risk leukemias and some genetic and metabolic disorders in children; however, comparative studies with other alternative donors are lacking in the literature.

64.4 Approaches to Improve Outcomes After UCBT

Strategies have been developed aiming to shorten the time to engraftment, improve immune reconstitution, and decrease NRM.

Approaches to improve outcomes after UCBT	Expert point of view
1. Refining criteria for UCB unit selection	Selection of the most appropriate UCB unit is key to transplant success. Recent comprehensive consensus guidelines have been recently published (Politikos et al. 2020; Ruggeri 2019). Selection of double cord blood units including high-resolution HLA typing for adults have been recently published (Fatobene et al. 2020; Fatobene et al. 2023). See chapter of banking, processing, and procurement of cord blood cells
2. Optimization of conditioning regimens	Specific conditioning regimen can influence transplant outcomes. See Sect. 64.5
3. Strategies aiming to shorten the time to engraftment:	
(a) Double UCBT	In children, two randomized trials have demonstrated no benefit and increased risk of GVHD (Wagner et al. 2014; Michel et al. 2016) In adults, retrospective studies showed no advantage when single-unit with TNC dose >2.5 × 10^7/kg available (Scaradavou et al. 2013)
(b) Co-infusion with third-party cells	Has consistently demonstrated benefit to accelerate hematopoietic recovery. No proved benefit on NRM or survival (Sanz et al. 2017)
(c) Ex vivo expansion of UCB cells	Omidubicel, the first FDA-approved UCB ex vivo-expanded cellular therapy product, has demonstrated faster engraftment, fewer infections, and decreased NRM compared with unmanipulated UCB that was confirmed in long-term follow-up (Horwitz et al. 2021). Should be considered the standard of care. However, logistical issues and costs may hamper real-life use. Other compounds to expand CB cells such as UM171 are currently being explored in phase 2 clinical trials with encouraging results (Cohen et al. 2020)
4. Improvement of supportive measures	Supportive care to prevent or treat opportunistic infections until neutrophil and immune recovery has occurred is critical in UCBT. See Sect. 64.7

64.5 Conditioning Regimens

The selection of conditioning regimen for HCT, including UCBT, should take into account the risk of toxicity and the risk of graft failure and relapse in malignant diseases. In UCBT, given the relatively lower cell dose (T cells and CD34+ cells) and the use of HLA-mismatched grafts, graft failure is of particular concern, especially in adults. It usually combines agents with additive immunosuppressive properties. The choice of the conditioning regimen is of primary importance and can influence transplant outcomes. A comprehensive and exhaustive review of myeloablative and nonmyeloblative/reduced intensity conditioning regimens in the UCBT setting has been published (Cord blood transplantations 2017). The American Society for Transplantation and Cellular Therapy Cord Blood Special Interest Group has also recently published their recommendations (Metheny et al. 2021).

The use of in vivo T-cell depletion with ATG in the conditioning regimen has been used to enhance myeloid engraftment as well as to prevent GVHD. Its use has been associated with reduced rates of GVHD but also with increased mortality due to delayed T-cell recovery in hematologic malignancies and is generally not recommended in this setting. Some data suggest that safety of ATG can be improved by adjusting timing or dose with ATG pharmacokinetics (Admiraal et al. 2016) to avoid posttransplantation exposure.

Some conditioning regimens options of varying intensities to be considered are as follows:

Myeloablative conditioning regimens (MAC)	
Chemotherapy-based	
• TBF regimen (Sanz et al. 2012)	TT 10 mg/kg + BU IV 9.6 mg/kg + Flu 150 mg/m^2
TBI-based	
• TBI-Flu-Cy regimen (Barker et al. 2015)	TBI 13.2 Gy + CY 120 mg/kg + Flu 75 mg/m^2
TT thiotepa, *BU* busulfan, *Flu* fludarabine, *TBI* total body irradiation, *CY* cyclophosphamide	
Intermediate intensity conditioning regimens (IIC)	
• TBI-TT-flu-Cy regimen (Barker et al. 2020)	TT 10 mg/kg + CY 50 mg/kg + Flu 150 mg/m^2 + TBI 4 Gy
TT thiotepa, Flu fludarabine, *CY* cyclophosphamide, *TBI* total body irradiation, *CY* cyclophosphamide	
Reduced intensity conditioning regimens (RIC)	
• rTBI-Ffu-Cy regimen (Fuchs et al. 2020)	TBI 2 Gy + CY 50 mg/kg + Flu 200 mg/m^2
TBI total body irradiation, *CY* cyclophosphamide, *Flu* fludarabine	

64.6 GVHD Prophylaxis

The most important advantage of UCB over unrelated donor grafts is the capability to tolerate HLA disparities and facilitate a low incidence of chronic GVHD. However, acute GVHD is still one of the most important contributors to morbidity and mortality. Different GVHD prophylaxis regimens have been explored with no evidence of benefit of any specific strategy. Methotrexate is generally not recommended to avoid myelotoxicity and delayed neutrophil recovery although it is widely used in Asia. The most frequently used regimen worldwide is the combination of calcineurin inhibitors for 6–9 months with mycophenolate mofetil for 2–6 months.

64.7 Supportive Care

The supportive measures described below are not intended to be recommendations but only to be taken into account and to consider their use in the context of each institution's own experience and epidemiology. The most common measures are described merely as a guide since they have a very variable level of evidence.

Prophylaxis, monitoring, and treatment options to be considered for infections.

Supportive measures for bacterial infections		
Prophylaxis	**Monitoring**	**Treatment**
Levofloxacin or ciprofloxacin	Surveillance cultures to detect colonization with multidrug-resistant gram-negative bacteria	Empirical antibacterial therapy according to institutional epidemiologic patterns
Supportive measures for fungal infections		
Prophylaxis	**Monitoring**	**Treatment**
Mold-covering azole	Galactomannan and beta-D-glucan assays[a]	Liposomal amphotericin B, azoles, and/or echinocandins (according to previous prophylaxis)
Supportive measures for viral infections		
Prophylaxis	**Monitoring** (quantitative PCR)	**Treatment**
CMV: letermovir	Weekly on days 0–100 and then as clinically indicated	Ganciclovir, valganciclovir, foscarnet
HHV-6: none	As clinically indicated	Ganciclovir, valganciclovir, foscarnet
Adenovirus: none	Weekly on days 0–100 and then as clinically indicated[b]	Cidofovir
EBV: none	Weekly on days 0–100 and then as clinically indicated	Preemptive rituximab
Supportive measures for protozoal infections		
Prophylaxis	**Monitoring**	**Treatment**
Pneumocystis: cotrimoxazol, pentamidine, or atovaquone	–	Cotrimoxazol, pentamidine, or atovaquone
Toxoplasmosis: cotrimoxazol, atovaquone, or pyrimethamine	–	Cotrimoxazol, atovaquone, or pyrimethamine

[a] Both have been included as microbiological criteria in the definitions of invasive fungal infections by the European Organization for Research and Treatment of Cancer (EORTC) and the Mycoses Study Group (MSG)
[b] Specially in children
[c] Reduced intensity conditioning and ATG are risk factors for EBV-PTLD

64.8 Results

UCBT outcomes are still improving in more recent years, probably explained by better patient and CBU selection, improved conditioning, and supported care. Registry data also showed important center effect with superior survival obtained in experienced centers. Eurocord recently updated clinical results.

Multiple retrospective studies have demonstrated that UCBT offers similar long-term outcomes compared with the gold standard of HLA-matched unrelated donor transplants in patients with hematologic malignancies, both in children and adults (Eapen et al. 2007; Brunstein et al. 2010; Atsuta et al. 2012). Interestingly, UCBT seems to offer a potent antileukemic efficacy, through yet unknown mechanisms. A markedly reduced relapse rate after UCBT as compared to URD transplantation in patients transplanted with minimal residual disease has been reported and needs to be validated (Milano et al, 2016). Moreover recently, the same observation has been reported in children with very high-risk leukemia transplanted with single CB units. Two-year EFS was 80% for CR 2, 67% for high risk (primary refractory or relapsed leukemia), and 61% for patients transplanted in CR1 (Horgan et al. 2023). In the same direction, very impressive results have been reported in children with genetic and metabolic disorders. Two-year overall survival was 91% with a median follow-up time of 4 years (Martinez et al. 2023).

64.9 Adults

	2-year OS (%)
Outcomes according to DRI	
Low	55 ± 3
Intermediate	47 ± 1
High	27 ± 2
Very high	19 ± 3
Disease-specific outcomes	
Acute leukemia	37 ± 1
MDS/MPS	32 ± 2
Lymphoproliferative disorders	45 ± 2
Plasma cell disorder	37 ± 5

OS overall survival, *DRI* disease risk index, *MDS* myelodysplastic syndrome, *MPS* myeloproliferative syndrome

64.10 Children

	2-year OS (%)
Malignant disorders	49 ± 1
Acute leukemia	52 ± 1
MDS	55 ± 3
Lymphoproliferative disorders	55 ± 3
Nonmalignant disorders	63 ± 1
Inborn error of metabolism	70 ± 2
Hemoglobinopathies	68 ± 9
Primary immunodeficiency	68 ± 2
Histiocytic disorders	60 ± 4
Bone marrow failure syndrome	52 ± 3

OS overall survival, *MDS* myelodysplastic syndrome

64.11 Conclusions

UCBT activity has drastically decreased in favor of other donor/stem cell options, specially haplo-HCT that shares the main advantage of UCB: high and rapid availability. However, UCB remains a valuable source of stem cells for HCT in selected patients that lack other options or in highly specialized units with preference of UCB. Important improvements in the field have taken place with a beneficial impact on NRM. Strategies to enhance engraftment with UCB ex vivo expansion have shown impressive success, although its widespread use and benefit in real-life clinical practice remain to be seen. There is still a room for improvement; however, clinical research in the field will be challenging in the future.

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